

Assessment Questions Python pandas numpy Matplotlib seaborn

Level 1 – Python + Pandas (1–5)

1.

1. Employee Salary Analysis

Dataset Columns:

Employee_ID, Name, Department, Age, Salary, Experience,

Tasks:

Highest salary employee name?

Average salary by department.

Employees with experience > 5 years.

Top 5 highest salaries.

Department-wise employee count.

```
import pandas as pd
data = {
    "Employee_ID": [101, 102, 103, 104, 105, 106, 107],
    "Name": ["Vijay", "Sanjay", "John", "Asuna", "Varshini", "Kirito", "Sahana"],
    "Department": ["HR", "IT", "IT", "Finance", "HR", "IT", "Finance"],
    "Age": [25, 30, 28, 35, 40, 29, 45],
    "Salary": [30000, 60000, 55000, 70000, 40000, 65000, 80000],
    "Experience": [2, 6, 4, 10, 8, 7, 12]
}
df = pd.DataFrame(data)
print(df.loc[df["Salary"].idxmax()])
print(df.groupby("Department")["Salary"].mean())
print(df[df["Experience"] > 5])
print(df.nlargest(5, "Salary"))
print(df["Department"].value_counts())
```

```

... Employee_ID      107
   Name      sahana
   Department  Finance
   Age      45
   Salary      80000
   Experience  12
   Name: 6, dtype: object
   Department
   Finance    75000.0
   HR         35000.0
   IT         60000.0
   Name: Salary, dtype: float64
      Employee_ID      Name Department  Age  Salary  Experience
1         102      Sanjay         IT    30   60000           6
3         104      asuna    Finance    35   70000          10
4         105  varshini         HR    40   40000           8
5         106      kirito         IT    29   65000           7
6         107      sahana    Finance    45   80000          12
      Employee_ID      Name Department  Age  Salary  Experience
6         107      sahana    Finance    45   80000          12
3         104      asuna    Finance    35   70000          10
5         106      kirito         IT    29   65000           7
1         102      Sanjay         IT    30   60000           6
2         103       john         IT    28   55000           4
   Department
   IT      3
   HR      2
   Finance  2
   Name: count, dtype: int64

```

2.

2. Student Marks Analysis

Columns:

Student, Math, Science, English, Attendance,

Tasks:

Total marks, Percentage, Grade (A/B/C/D), Topper, Students below 40%

```
import pandas as pd
data = {
    "Student":["Vijay","sanjay","john","Sahana","Varshini"],
    "Math":[85,45,92,30,60],
    "Science":[90,40,88,35,70],
    "English":[80,50,95,25,65],
    "Attendance":[95,85,98,75,90]
}
df = pd.DataFrame(data)
df["Total"]=df["Math"]+df["Science"]+df["English"]
print(df["Total"])
df["Percentage"] =(df["Total"]/300)*100
print(df["Percentage"] )
df["Grade"] ="D"
df.loc[df["Percentage"]>=40,"Grade"]="C"
df.loc[df["Percentage"]>=60,"Grade"]="B"
df.loc[df["Percentage"]>=80,"Grade"]="A"
print(df)
print(df.loc[df["Total"].idxmax()])
print(df[df["Percentage"]<40])
```

```

... 0    255
    1    135
    2    275
    3     90
    4    195
    Name: Total, dtype: int64
    0    85.000000
    1    45.000000
    2    91.666667
    3    30.000000
    4    65.000000
    Name: Percentage, dtype: float64
      Student  Math  Science  English  Attendance  Total  Percentage  Grade
0    Vijay    85     90     80         95      255    85.000000    A
1   sanjay    45     40     50         85      135    45.000000    C
2     john    92     88     95         98      275    91.666667    A
3   Sahana    30     35     25         75       90    30.000000    D
4  Varshini    60     70     65         90      195    65.000000    B
Student              john
Math                  92
Science               88
English               95
Attendance            98
Total                 275
Percentage    91.666667
Grade          A
Name: 2, dtype: object
      Student  Math  Science  English  Attendance  Total  Percentage  Grade
3   Sahana    30     35     25         75       90     30.0      D

```

3.

3. Hospital Patient Data

Columns:

Patient_ID, Age, Gender, Disease, Bill, Days_Admitted,

Tasks:

Average bill, Disease-wise patient count, Highest bill, Patients admitted >7 days, Male vs Female count.

```

import pandas as pd
data = {
    "Patient_ID": [101, 102, 103, 104, 105],
    "Age": [20, 21, 25, 49, 33],
    "Gender": ["Male", "Female", "Male", "Female", "Male"],
    "Disease": ["Fever", "Diabetes", "Fever", "Heart", "Cancer"],
    "Bill": [15000, 45000, 18000, 70000, 55000],
    "DaysAdmitted": [3, 10, 5, 12, 8]
}
df = pd.DataFrame(data)
print(df["Bill"].mean())
print(df["Disease"].value_counts())
print(df.loc[df["Bill"].idxmax()])
print(df[df["DaysAdmitted"] > 7])
print(df["Gender"].value_counts())

```

```

... 40600.0
Disease
Fever      2
Diabetes    1
Heart       1
Cancer      1
Name: count, dtype: int64
Patient_ID      104
Age              49
Gender          Female
Disease          Heart
Bill            70000
DaysAdmitted     12
Name: 3, dtype: object
   Patient_ID  Age  Gender  Disease  Bill  DaysAdmitted
1         102   21  Female  Diabetes  45000             10
3         104   49  Female    Heart  70000             12
4         105   33   Male    Cancer  55000              8
Gender
Male      3
Female    2
Name: count, dtype: int64

```

4.

4. Supermarket Sales

Columns:

Invoice, Product, Category, Quantity, Price, Date,

Tasks:

Total revenue, Best selling product, Category-wise sales, Average order value, Top 10 invoices

```
import pandas as pd
data = {
    "Invoice": [1001, 1002, 1003, 1004, 1005, 1006],
    "Product": ["Rice", "Milk", "Sugar", "Rice", "Oil", "Milk"],
    "Category": ["Grocery", "Dairy", "Grocery", "Grocery", "Cooking", "Dairy"],
    "Quantity": [2, 5, 3, 4, 2, 6],
    "Price": [500, 50, 60, 500, 200, 50],
    "Date": ["2026-08-01", "2026-08-01", "2026-08-02",
             "2026-08-02", "2026-08-03", "2026-08-03"]
}
df = pd.DataFrame(data)
df["Revenue"] = df["Quantity"] * df["Price"]
print(df["Revenue"].sum())
print(df.groupby("Product")["Quantity"].sum().idxmax())
print(df.groupby("Category")["Revenue"].sum())
print(df["Revenue"].mean())
print(df.nlargest(10, "Revenue"))
```

```
... 4130
Milk
Category
Cooking      400
Dairy        550
Grocery     3180
Name: Revenue, dtype: int64
688.33333333333334
```

	Invoice	Product	Category	Quantity	Price	Date	Revenue
3	1004	Rice	Grocery	4	500	2026-08-02	2000
0	1001	Rice	Grocery	2	500	2026-08-01	1000
4	1005	Oil	Cooking	2	200	2026-08-03	400
5	1006	Milk	Dairy	6	50	2026-08-03	300
1	1002	Milk	Dairy	5	50	2026-08-01	250
2	1003	Sugar	Grocery	3	60	2026-08-02	180

5.

5. Online Food Orders

Columns:

Order_ID, Restaurant, Customer, Amount, Rating, Delivery_Time,

Tasks:

Highest rated restaurant, Average delivery time, Orders above ₹1000, Restaurant-wise revenue, Rating distribution

```
import pandas as pd

data = {
    "Order_ID": [101, 102, 103, 104, 105, 106],
    "Restaurant": ["KFC", "Dominos", "KFC", "Pizza Hut", "Dominos", "Burger King"],
    "Customer": ["Vijay", "John", "Sanjay", "Sahana", "Varshini", "Asuna"],
    "Amount": [850, 1200, 950, 1500, 700, 1100],
    "Rating": [4.5, 4.8, 4.2, 4.9, 4.6, 4.4],
    "DeliveryTime": [30, 45, 25, 50, 35, 40]
}

df = pd.DataFrame(data)
print(df.loc[df["Rating"].idxmax()])
print(df["DeliveryTime"].mean())
print(df[df["Amount"] > 1000])
print(df.groupby("Restaurant")["Amount"].sum())
print(df["Rating"].value_counts().sort_index())
```

```
... Order_ID      104
   Restaurant    Pizza Hut
   Customer      Sahana
   Amount        1500
   Rating         4.9
   DeliveryTime    50
   Name: 3, dtype: object
37.5
   Order_ID  Restaurant  Customer  Amount  Rating  DeliveryTime
1       102     Dominos    John    1200    4.8           45
3       104     Pizza Hut  Sahana    1500    4.9           50
5       106  Burger King   Asuna    1100    4.4           40
Restaurant
Burger King    1100
Dominos        1900
KFC            1800
Pizza Hut      1500
Name: Amount, dtype: int64
Rating
4.2    1
4.4    1
4.5    1
4.6    1
4.8    1
4.9    1
Name: count, dtype: int64
```

Level 2 – NumPy + Pandas (6–10)

6.

Level 2 – NumPy + Pandas (6–10)

6. Weather Analysis

Columns:

Date, Temperature, Humidity, Rainfall, WindSpeed,

Tasks:

Hottest day, Coldest day, Average temperature, Days above 35°C, Correlation between temperature & humidity.

```
import numpy as np
import pandas as pd
data = {
    "Date": ["2026-08-01", "2026-08-02", "2026-08-03",
            "2026-08-04", "2026-08-05", "2026-08-06"],
    "Temperature": [32, 38, 36, 29, 41, 35],
    "Humidity": [75, 60, 65, 80, 55, 70],
    "Rainfall": [5, 0, 2, 10, 0, 1],
    "WindSpeed": [12, 15, 18, 10, 20, 16]
}
df = pd.DataFrame(data)
print(df.loc[df["Temperature"].idxmax()])
print(df.loc[df["Temperature"].idxmin()])
print(np.mean(df["Temperature"]))
print(df[df["Temperature"] > 35])
print(df["Temperature"].corr(df["Humidity"]))
```

```
...   Date      2026-08-05
      Temperature      41
      Humidity        55
      Rainfall         0
      WindSpeed       20
      Name: 4, dtype: object
      Date      2026-08-04
      Temperature      29
      Humidity        80
      Rainfall        10
      WindSpeed       10
      Name: 3, dtype: object
      35.166666666666664
      Date      Temperature      Humidity      Rainfall      WindSpeed
1  2026-08-02           38           60           0           15
2  2026-08-03           36           65           2           18
4  2026-08-05           41           55           0           20
      -0.990730299762568
```

7.

7. IPL Dataset

Columns:

Player, Team, Runs, Matches, StrikeRate, Fours, Sixes,

Tasks:

Orange cap, Average strike rate, Team-wise runs, Top 10 batsmen, Highest six hitter

```
import numpy as np
import pandas as pd
data = {
    "Player": ["Vijay", "John", "Sanjay", "Sahana", "Varshini", "Asuna"],
    "Team": ["RCB", "MI", "GT", "LSG", "MI", "CSK"],
    "Runs": [741, 520, 690, 610, 480, 350],
    "Matches": [15, 14, 15, 14, 13, 14],
    "StrikeRate": [154.7, 145.3, 151.8, 142.5, 170.2, 165.5],
    "Fours": [62, 45, 58, 50, 38, 20],
    "Sixes": [38, 28, 32, 25, 30, 24]
}
df = pd.DataFrame(data)
print(df.loc[df["Runs"].idxmax()])
print(np.mean(df["StrikeRate"]))
print(df.groupby("Team")["Runs"].sum())
print(df.nlargest(10, "Runs"))
print(df.loc[df["Sixes"].idxmax()])
```

```
... Player      Vijay
Team          RCB
Runs          741
Matches        15
StrikeRate    154.7
Fours         62
Sixes         38
Name: 0, dtype: object
155.0
Team
CSK      350
GT       690
LSG      610
MI     1000
RCB      741
Name: Runs, dtype: int64
   Player Team  Runs  Matches  StrikeRate  Fours  Sixes
0  Vijay  RCB   741      15      154.7     62    38
2  Sanjay  GT   690      15      151.8     58    32
3  Sahana LSG   610      14      142.5     50    25
1   John   MI   520      14      145.3     45    28
4  Varshini MI   480      13      170.2     38    30
5   Asuna  CSK   350      14      165.5     20    24
Player      Vijay
Team          RCB
Runs          741
Matches        15
StrikeRate    154.7
Fours         62
Sixes         38
Name: 0, dtype: object
```

8.

8. Bank Customer Data

Columns:

Customer, Balance, Age, Loan, CreditScore,

Tasks:

Customers with balance >1 lakh, Average credit score, Loan holders percentage, Highest balance, Age distribution

```
import numpy as np
import pandas as pd
data = {
    "Customer": ["Vijay", "John", "Sanjay", "Sahana", "Varshini", "David"],
    "Balance": [150000, 85000, 220000, 98000, 175000, 120000],
    "Age": [25, 40, 35, 28, 32, 45],
    "Loan": ["Yes", "No", "Yes", "No", "Yes", "No"],
    "CreditScore": [780, 650, 810, 720, 790, 690]
}
df = pd.DataFrame(data)
print(df[df["Balance"] > 100000])
print(np.mean(df["CreditScore"]))
loanpercentage = (df["Loan"] == "Yes").mean() * 100
print(loanpercentage)
print(df.loc[df["Balance"].idxmax()])
print(df["Age"].value_counts().sort_index())
```

```
...      Customer  Balance  Age  Loan  CreditScore
0      Vijay    150000    25   Yes         780
2      Sanjay    220000    35   Yes         810
4  Varshini    175000    32   Yes         790
5      David    120000    45   No         690
740.0
50.0
Customer      Sanjay
Balance      220000
Age           35
Loan          Yes
CreditScore   810
Name: 2, dtype: object
Age
25    1
28    1
32    1
35    1
40    1
45    1
Name: count, dtype: int64
```

9.

9. Movie Dataset

Columns:

Movie, Genre, Budget, Collection, Rating,

Tasks:

Profit, Highest collection, Genre-wise average rating, ROI, Movies with loss.

```
import numpy as np
import pandas as pd
data = {
    "Movie": ["Leo", "Jailer", "Kalki", "KGF", "Pushpa", "Master"],
    "Genre": ["Action", "Action", "Sci-Fi", "Action", "Drama", "Action"],
    "Budget": [250, 200, 350, 100, 180, 150],
    "Collection": [620, 650, 1100, 1200, 150, 300],
    "Rating": [8.2, 8.5, 9.0, 8.8, 7.5, 8.0]
}
df = pd.DataFrame(data)
df["Profit"] = df["Collection"] - df["Budget"]
print(df[["Movie", "Profit"]])
print(df.loc[df["Collection"].idxmax()])
print(df.groupby("Genre")["Rating"].mean())
df["ROI"] = (df["Profit"] / df["Budget"]) * 100
print(df[["Movie", "ROI"]])
print(df[df["Profit"] < 0])
```

```
...
   Movie  Profit
0    Leo    370
1  Jailer    450
2   Kalki    750
3    KGF   1100
4  Pushpa   -30
5  Master    150
Movie    KGF
Genre    Action
Budget      100
Collection  1200
Rating      8.8
Profit     1100
Name: 3, dtype: object
Genre
Action    8.375
Drama     7.500
Sci-Fi    9.000
Name: Rating, dtype: float64
   Movie  ROI
0    Leo  148.000000
1  Jailer  225.000000
2   Kalki  214.285714
3    KGF  1100.000000
4  Pushpa  -16.666667
5  Master  100.000000
   Movie  Genre  Budget  Collection  Rating  Profit  ROI
4  Pushpa  Drama    180         150     7.5     -30  -16.666667
```

10.

10. E-Commerce Orders

Columns:

Order_ID, Customer, Product, Price, Quantity, Discount,

Tasks:

Final amount, Highest discount, Monthly sales, Most ordered product, Average cart value.

```
import numpy as np
import pandas as pd
data = {
    "Order_ID": [101, 102, 103, 104, 105, 106],
    "Customer": ["Vijay", "John", "Sanjay", "Sahana", "Varshini", "David"],
    "Product": ["Laptop", "Mobile", "Laptop", "Headphones", "Mobile", "Laptop"],
    "Price": [50000, 20000, 50000, 5000, 20000, 50000],
    "Quantity": [1, 2, 1, 3, 1, 2],
    "Discount": [5000, 2000, 3000, 500, 1000, 4000],
    "Month": ["Jan", "Jan", "Feb", "Feb", "Mar", "Mar"]
}
df = pd.DataFrame(data)
df["Final_Amount"] = (df["Price"] * df["Quantity"]) - df["Discount"]
print(df[["Order_ID", "Final_Amount"]])
print(df.loc[df["Discount"].idxmax()])
print(df.groupby("Month")["Final_Amount"].sum())
print(df.groupby("Product")["Quantity"].sum().idxmax())
print(np.mean(df["Final_Amount"]))
```

```
...      Order_ID  Final_Amount
0         101         45000
1         102         38000
2         103         47000
3         104         14500
4         105         19000
5         106         96000
Order_ID      101
Customer      Vijay
Product       Laptop
Price         50000
Quantity       1
Discount       5000
Month         Jan
Final_Amount  45000
Name: 0, dtype: object
Month
Feb      61500
Jan      83000
Mar     115000
Name: Final_Amount, dtype: int64
Laptop
43250.0
```

Level 3 – Matplotlib (11–15)

11.

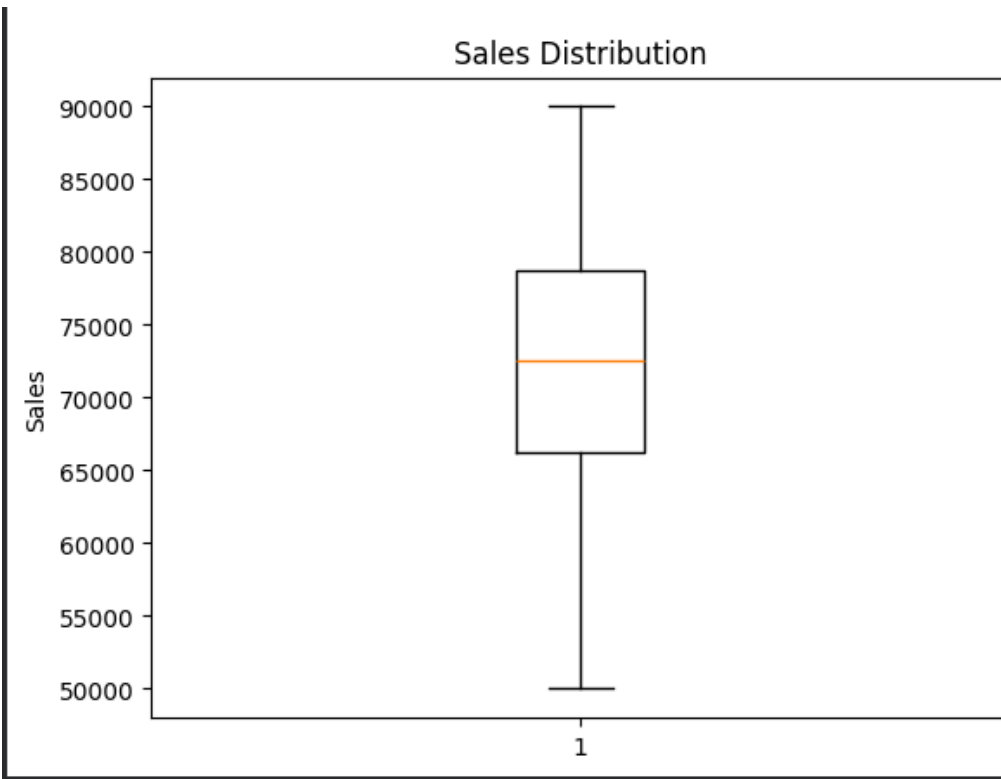
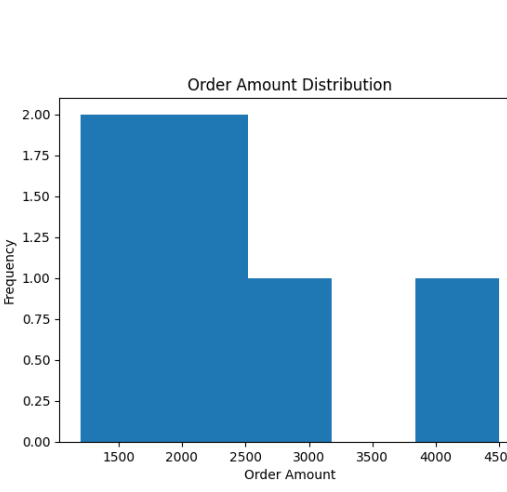
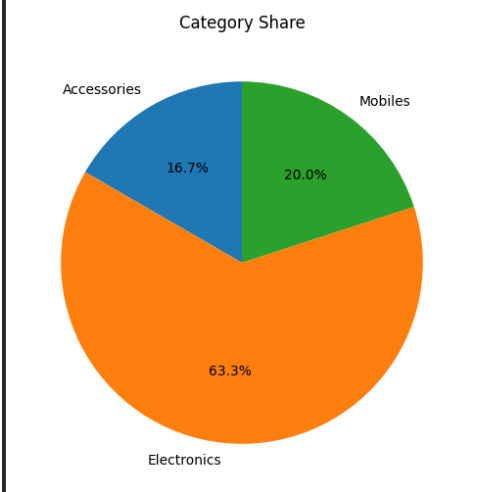
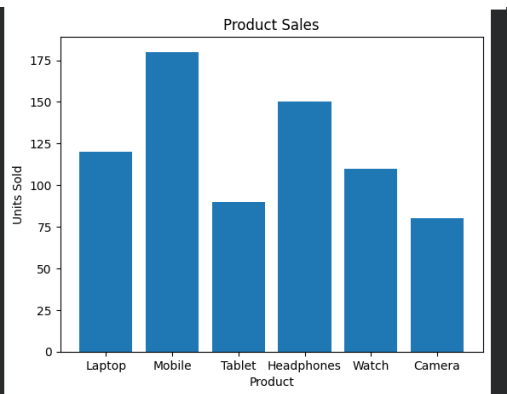
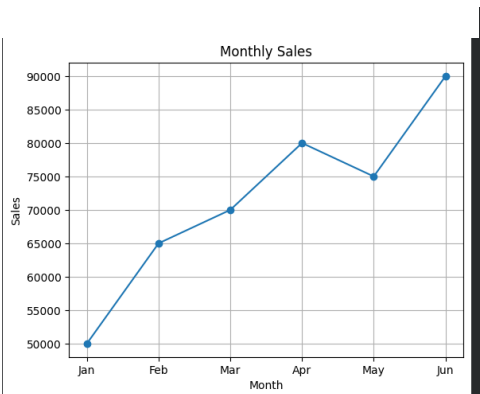
Level 3 – Matplotlib (11–15)

11. Sales Dashboard

Create:

Line Chart → Monthly Sales, Bar Chart → Product Sales, Pie Chart → Category Share, Histogram → Order Amount, Box Plot → Sales Distribution

```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [50000, 65000, 70000, 80000, 75000, 90000],
    "Product": ["Laptop", "Mobile", "Tablet", "Headphones", "Watch", "Camera"],
    "ProductSales": [120, 180, 90, 150, 110, 80],
    "Category": ["Electronics", "Mobiles", "Electronics", "Accessories", "Accessories", "Electronics"],
    "CategoryShare": [40, 30, 20, 10, 15, 35],
    "OrderAmount": [1200, 2500, 3000, 4500, 1800, 2200]
}
df = pd.DataFrame(data)
plt.plot(df["Month"], df["Sales"], marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid()
plt.show()
plt.bar(df["Product"], df["ProductSales"])
plt.title("Product Sales")
plt.xlabel("Product")
plt.ylabel("Units Sold")
plt.show()
category = df.groupby("Category")["CategoryShare"].sum()
plt.figure(figsize=(6, 6))
plt.pie(category.values, labels=category.index, autopct="%1.1f%%", startangle=90)
plt.title("Category Share")
plt.show()
plt.hist(data["OrderAmount"], bins=5)
plt.title("Order Amount Distribution")
plt.xlabel("Order Amount")
plt.ylabel("Frequency")
plt.show()
plt.boxplot(df["Sales"])
plt.title("Sales Distribution")
plt.ylabel("Sales")
plt.show()
```



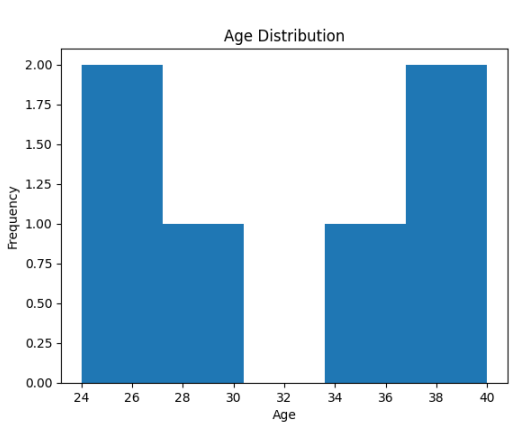
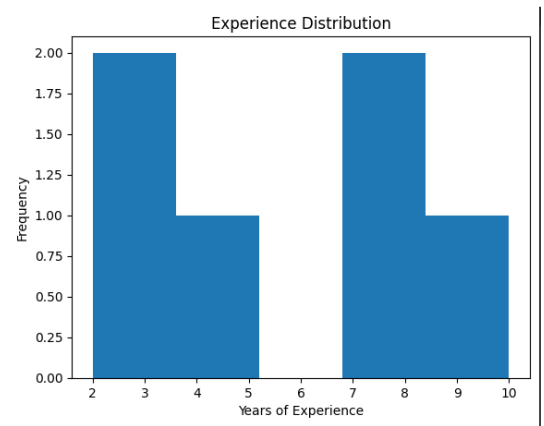
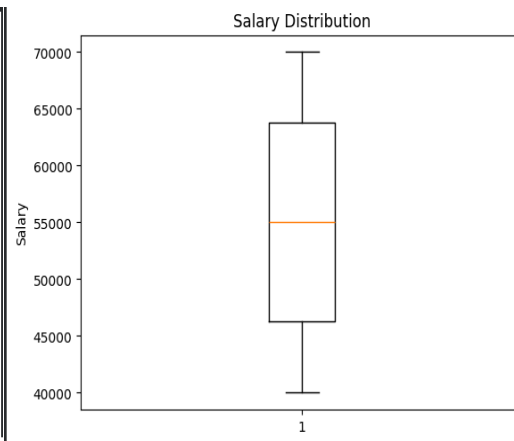
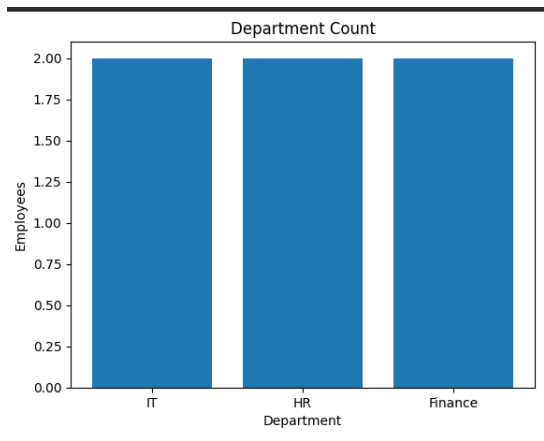
12.

12. Employee Dashboard

Visualize

Department count, Salary distribution, Experience distribution, Age histogram, Salary vs Experience scatter.

```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Employee": ["Vijay", "John", "Sanjay", "Sahana", "Varshini", "David"],
    "Department": ["IT", "HR", "Finance", "IT", "HR", "Finance"],
    "Salary": [50000, 40000, 60000, 70000, 45000, 65000],
    "Experience": [2, 5, 8, 10, 3, 7],
    "Age": [24, 30, 35, 40, 27, 38]
}
df = pd.DataFrame(data)
dept = df["Department"].value_counts()
plt.bar(dept.index, dept.values)
plt.title("Department Count")
plt.xlabel("Department")
plt.ylabel("Employees")
plt.show()
plt.boxplot(df["Salary"])
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
plt.hist(df["Experience"], bins=5)
plt.title("Experience Distribution")
plt.xlabel("Years of Experience")
plt.ylabel("Frequency")
plt.show()
plt.hist(df["Age"], bins=5)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
plt.scatter(df["Experience"], df["Salary"])
plt.title("Salary vs Experience")
plt.xlabel("Experience (Years)")
plt.ylabel("Salary")
plt.grid()
plt.show()
```



13.

13. Hospital Dashboard

Create

Disease count, Average bill, Gender ratio, Age distribution, Admission days histogram.

```
import matplotlib.pyplot as plt
import pandas as pd

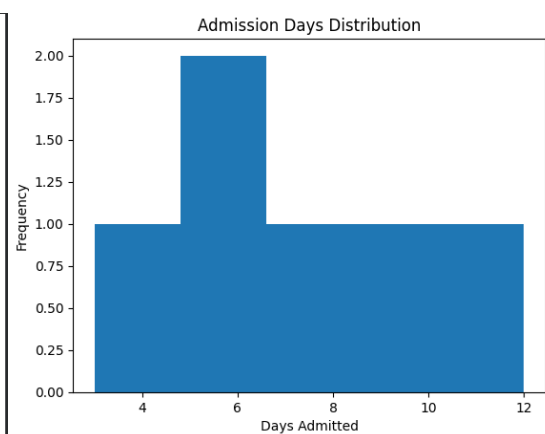
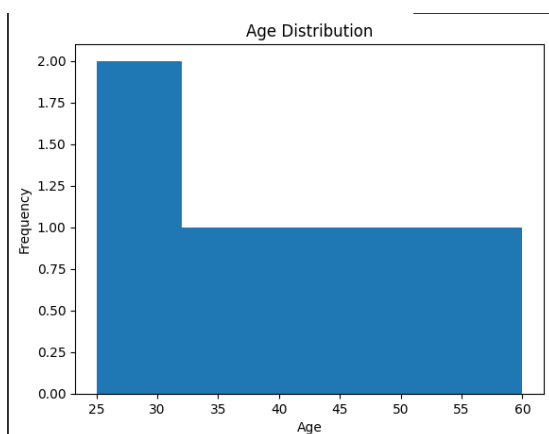
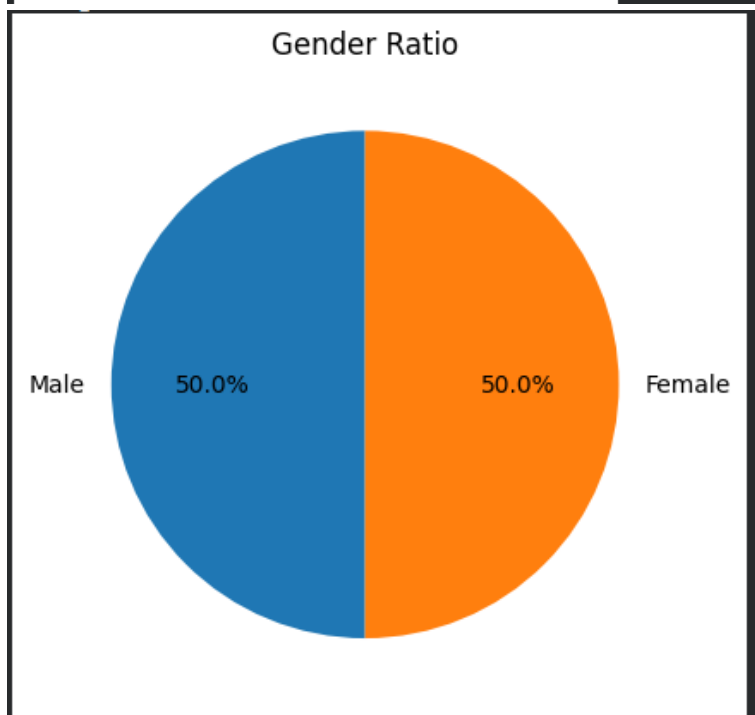
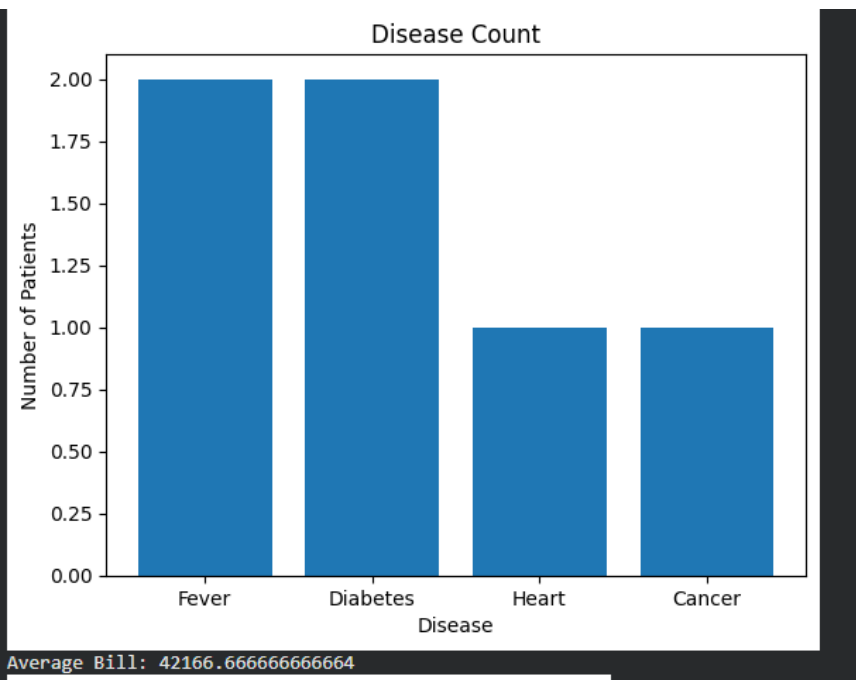
data = {
    "Patient_ID": [101, 102, 103, 104, 105, 106],
    "Age": [25, 40, 35, 50, 28, 60],
    "Gender": ["Male", "Female", "Male", "Female", "Male", "Female"],
    "Disease": ["Fever", "Diabetes", "Fever", "Heart", "Cancer", "Diabetes"],
    "Bill": [15000, 45000, 18000, 70000, 55000, 50000],
    "DaysAdmitted": [3, 10, 5, 12, 8, 6]
}

df = pd.DataFrame(data)
disease = df["Disease"].value_counts()
plt.bar(disease.index, disease.values)
plt.title("Disease Count")
plt.xlabel("Disease")
plt.ylabel("Number of Patients")
plt.show()

avg_bill = df["Bill"].mean()
print("Average Bill:", avg_bill)
gender = df["Gender"].value_counts()
plt.pie(gender.values, labels=gender.index, autopct="%1.1f%%", startangle=90)
plt.title("Gender Ratio")
plt.show()

plt.hist(df["Age"], bins=5)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()

plt.hist(df["DaysAdmitted"], bins=5)
plt.title("Admission Days Distribution")
plt.xlabel("Days Admitted")
plt.ylabel("Frequency")
plt.show()
```



14.

14. IPL Dashboard

Create

Top 10 run scorers, Team total runs, Strike rate histogram, Runs vs Matches, Sixes comparison.

```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Player": ["Virat", "Rohit", "Gill", "KL Rahul", "Surya", "Dhoni"],
    "Team": ["RCB", "MI", "GT", "LSG", "MI", "CSK"],
    "Runs": [741, 520, 690, 610, 480, 350],
    "Matches": [15, 14, 15, 14, 13, 14],
    "StrikeRate": [154.7, 145.3, 151.8, 142.5, 170.2, 165.5],
    "Fours": [62, 45, 58, 50, 38, 20],
    "Sixes": [38, 28, 32, 25, 30, 24]
}

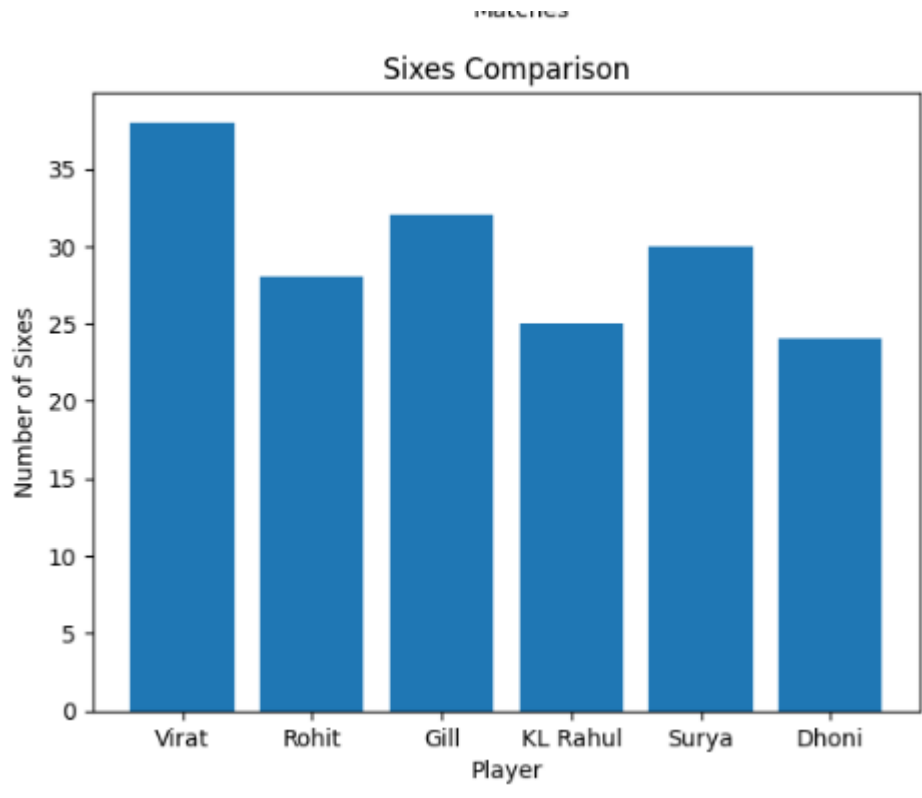
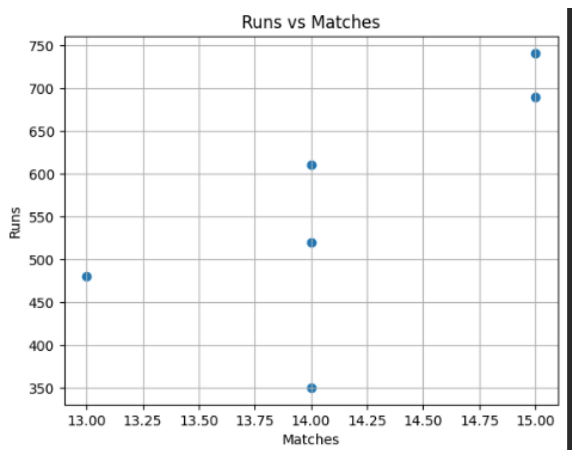
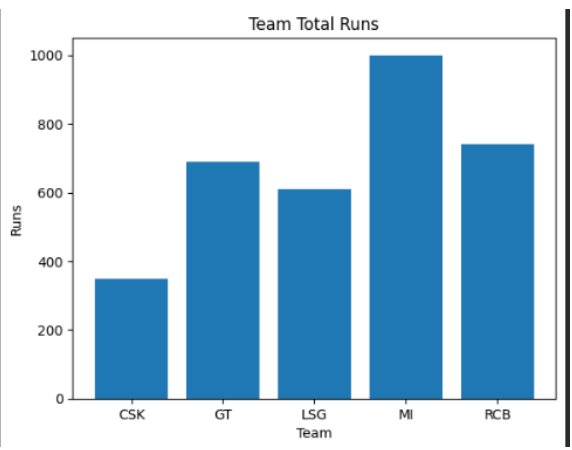
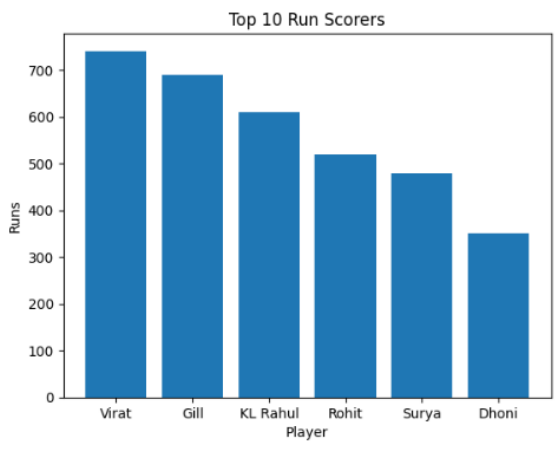
df = pd.DataFrame(data)
top10 = df.nlargest(10, "Runs")
plt.bar(top10["Player"], top10["Runs"])
plt.title("Top 10 Run Scorers")
plt.xlabel("Player")
plt.ylabel("Runs")
plt.show()

team_runs = df.groupby("Team")["Runs"].sum()
plt.bar(team_runs.index, team_runs.values)
plt.title("Team Total Runs")
plt.xlabel("Team")
plt.ylabel("Runs")
plt.show()

plt.hist(df["StrikeRate"], bins=5)
plt.title("Strike Rate Distribution")
plt.xlabel("Strike Rate")
plt.ylabel("Frequency")
plt.show()

plt.scatter(df["Matches"], df["Runs"])
plt.title("Runs vs Matches")
plt.xlabel("Matches")
plt.ylabel("Runs")
plt.grid()
plt.show()

plt.bar(df["Player"], df["Sixes"])
plt.title("Sixes Comparison")
plt.xlabel("Player")
plt.ylabel("Number of Sixes")
plt.show()
```



15.

15. Weather Dashboard

Visualize

Temperature trend, Rainfall trend, Humidity histogram, Wind speed box plot, Correlation scatter

```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Date": ["1-Aug", "2-Aug", "3-Aug", "4-Aug", "5-Aug", "6-Aug"],
    "Temperature": [32, 35, 38, 34, 36, 33],
    "Rainfall": [5, 0, 12, 8, 3, 10],
    "Humidity": [70, 65, 60, 75, 68, 72],
    "WindSpeed": [12, 15, 18, 14, 16, 13]
}

df = pd.DataFrame(data)

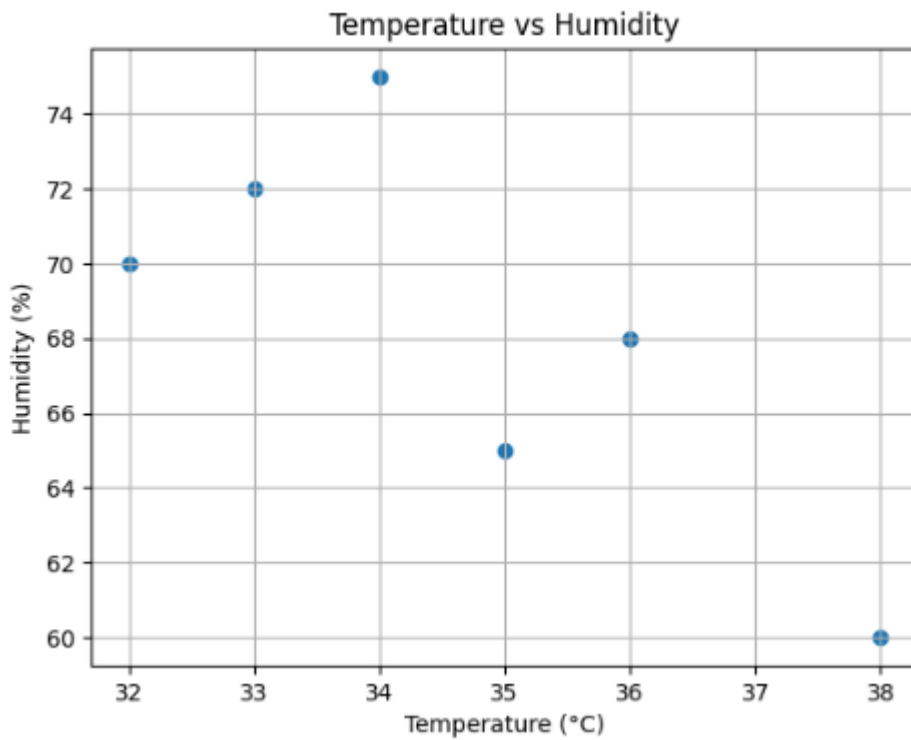
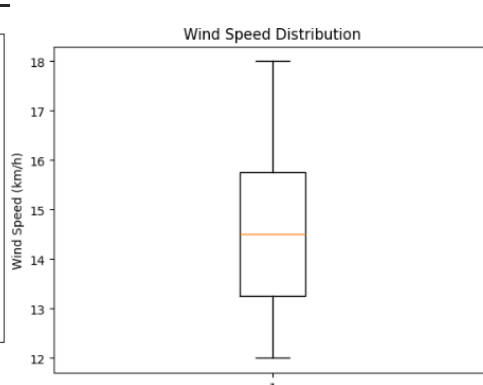
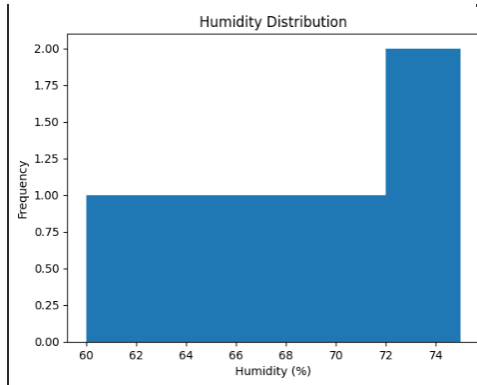
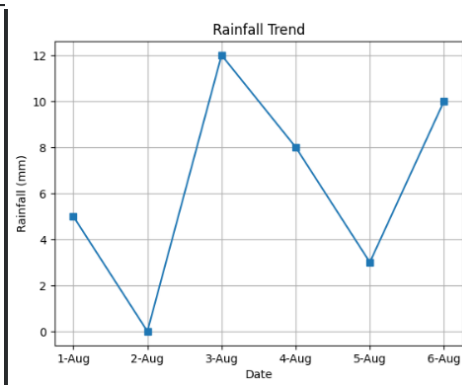
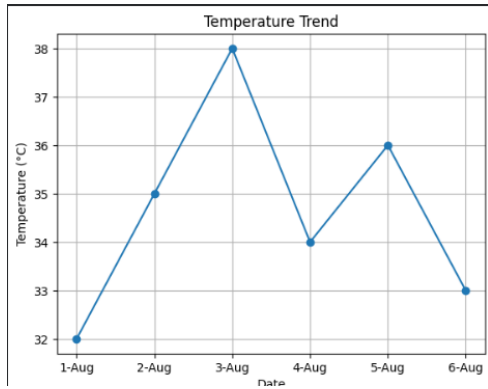
plt.plot(df["Date"], df["Temperature"], marker="o")
plt.title("Temperature Trend")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.grid(True)
plt.show()

plt.plot(df["Date"], df["Rainfall"], marker="s")
plt.title("Rainfall Trend")
plt.xlabel("Date")
plt.ylabel("Rainfall (mm)")
plt.grid(True)
plt.show()

plt.hist(df["Humidity"], bins=5)
plt.title("Humidity Distribution")
plt.xlabel("Humidity (%)")
plt.ylabel("Frequency")
plt.show()

plt.boxplot(df["WindSpeed"])
plt.title("Wind Speed Distribution")
plt.ylabel("Wind Speed (km/h)")
plt.show()

plt.scatter(df["Temperature"], df["Humidity"])
plt.title("Temperature vs Humidity")
plt.xlabel("Temperature (°C)")
plt.ylabel("Humidity (%)")
plt.grid()
plt.show()
```



Level 4 – Seaborn (16–20)

16.

16. Titanic Dataset

Perform

Countplot, Boxplot, Violinplot, Pairplot, Heatmap,

Questions

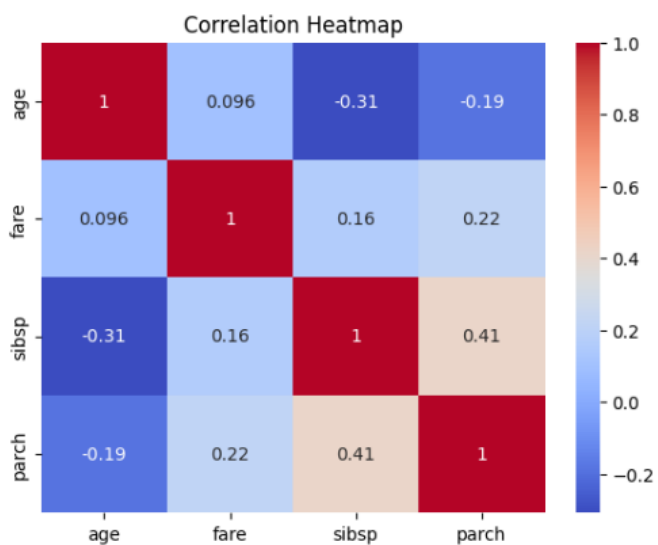
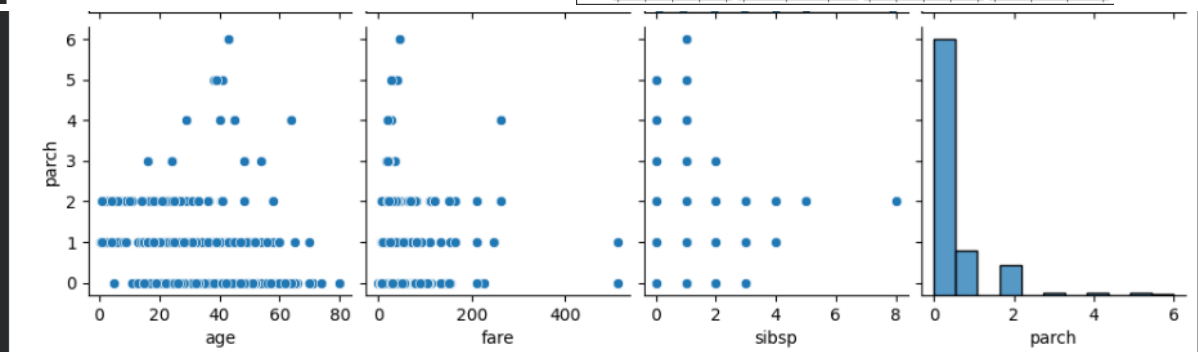
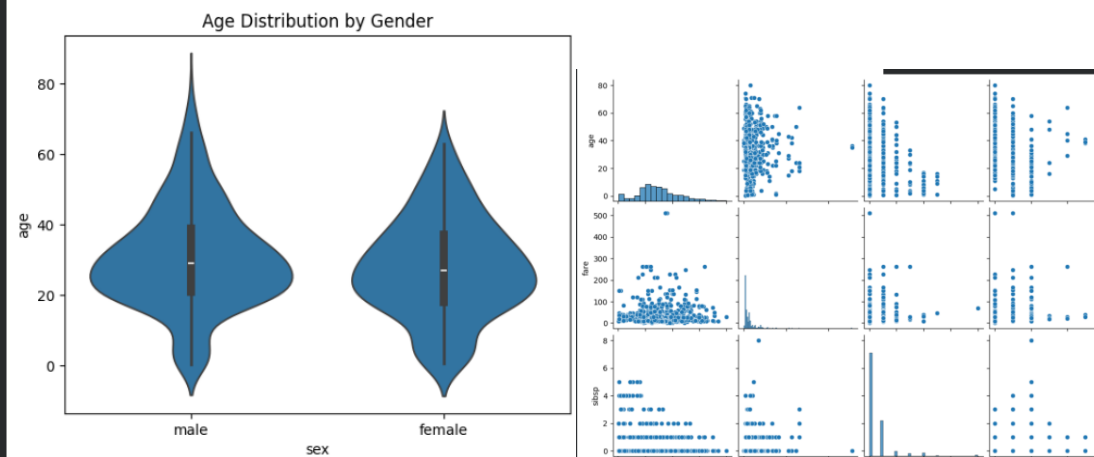
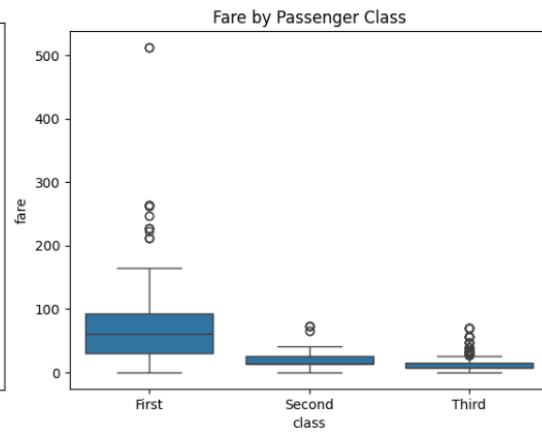
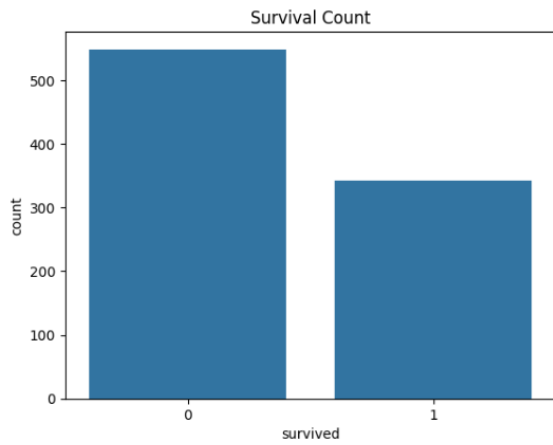
Survival by gender?, Survival by class?, Age distribution?, Fare distribution?.

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

df=sns.load_dataset("titanic")
print(df.head())
sns.countplot(data=df, x="survived")
plt.title("Survival Count")
plt.show()
sns.boxplot(data=df, x="class", y="fare")
plt.title("Fare by Passenger Class")
plt.show()
sns.violinplot(data=df, x="sex", y="age")
plt.title("Age Distribution by Gender")
plt.show()
sns.pairplot(df[["age", "fare", "sibsp", "parch"]])
plt.show()
sns.heatmap(df[["age", "fare", "sibsp", "parch"]].corr(),annot=True,cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

```
...   survived  pclass    sex  age  sibsp  parch   fare embarked  class \
0         0         3   male  22.0     1     0   7.2500         S   Third
1         1         1  female  38.0     1     0  71.2833         C   First
2         1         3  female  26.0     0     0   7.9250         S   Third
3         1         1  female  35.0     1     0  53.1000         S   First
4         0         3   male  35.0     0     0   8.0500         S   Third

   who  adult_male  deck  embark_town  alive  alone
0   man         True  NaN  Southampton    no  False
1 woman        False   C   Cherbourg   yes  False
2 woman        False  NaN  Southampton   yes   True
3 woman        False   C   Southampton   yes  False
4   man         True  NaN  Southampton    no   True
```



17.

17. Netflix Dataset

Columns

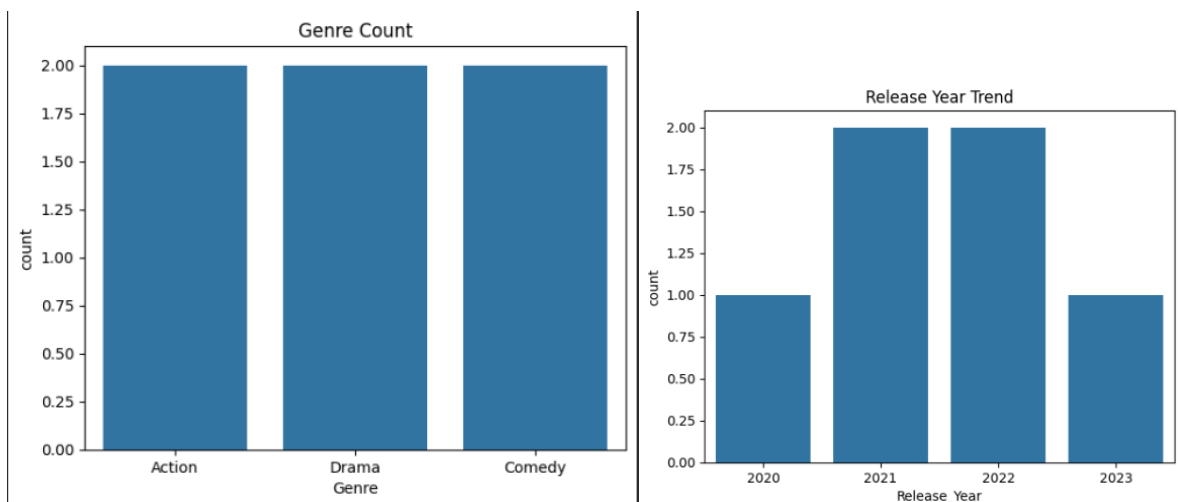
Title Genre Country Rating Duration Release_Year

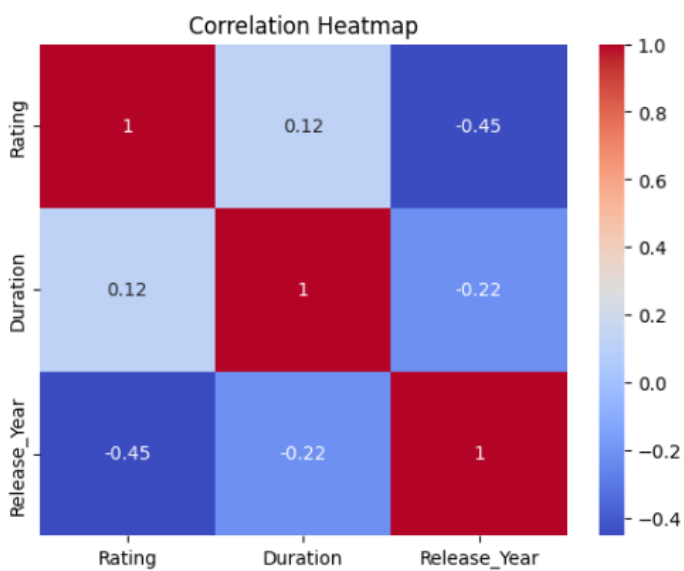
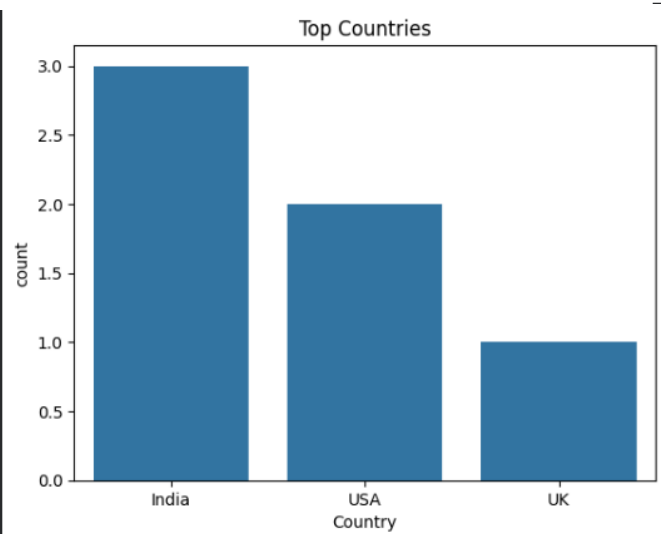
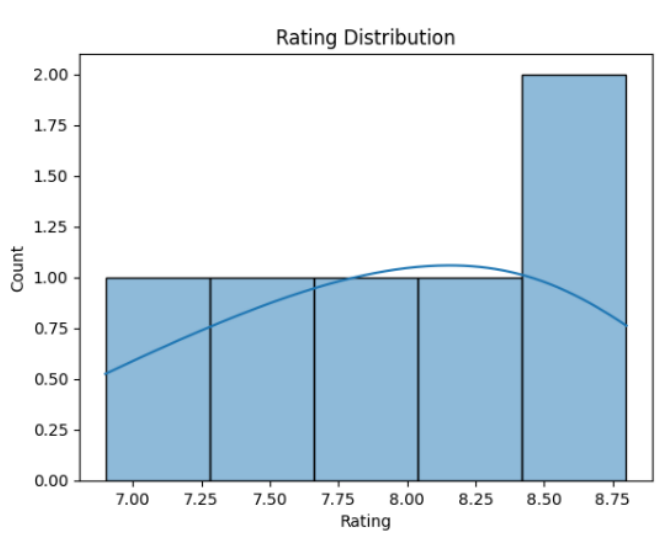
Tasks

Genre countplot Release year trend Rating distribution Top countries Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Title": ["Movie A", "Movie B", "Movie C", "Movie D", "Movie E", "Movie F"],
    "Genre": ["Action", "Drama", "Comedy", "Action", "Drama", "Comedy"],
    "Country": ["India", "USA", "India", "UK", "USA", "India"],
    "Rating": [8.5, 7.8, 6.9, 8.2, 7.5, 8.8],
    "Duration": [120, 140, 110, 150, 130, 115],
    "Release_Year": [2020, 2021, 2022, 2021, 2023, 2022]
}

df = pd.DataFrame(data)
sns.countplot(data=df, x="Genre")
plt.title("Genre Count")
plt.show()
sns.countplot(data=df, x="Release_Year")
plt.title("Release Year Trend")
plt.show()
sns.histplot(data=df, x="Rating", bins=5, kde=True)
plt.title("Rating Distribution")
plt.show()
sns.countplot(data=df, x="Country")
plt.title("Top Countries")
plt.show()
sns.heatmap(df[["Rating", "Duration", "Release_Year"]].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```





18.

18. Car Dataset

Columns

Brand Price Mileage Horsepower FuelType

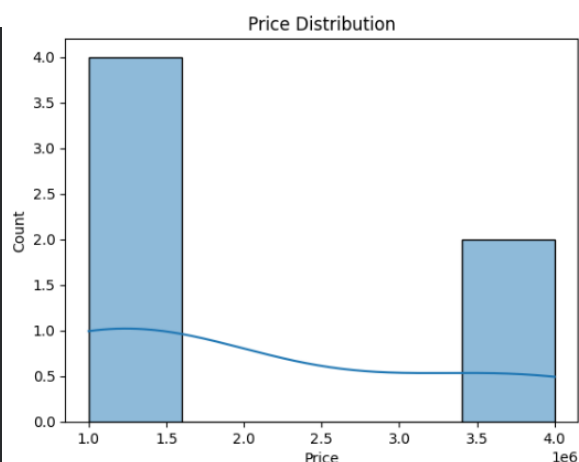
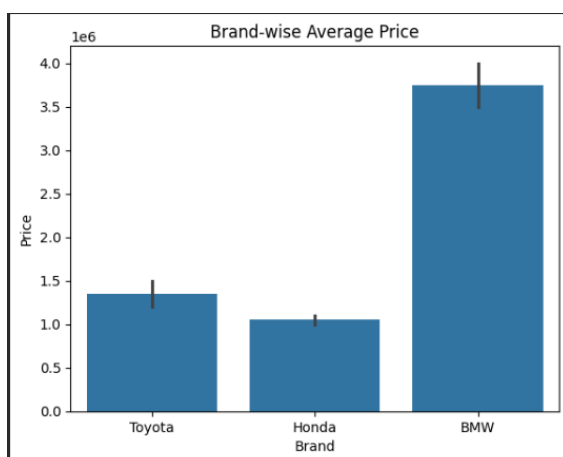
Tasks

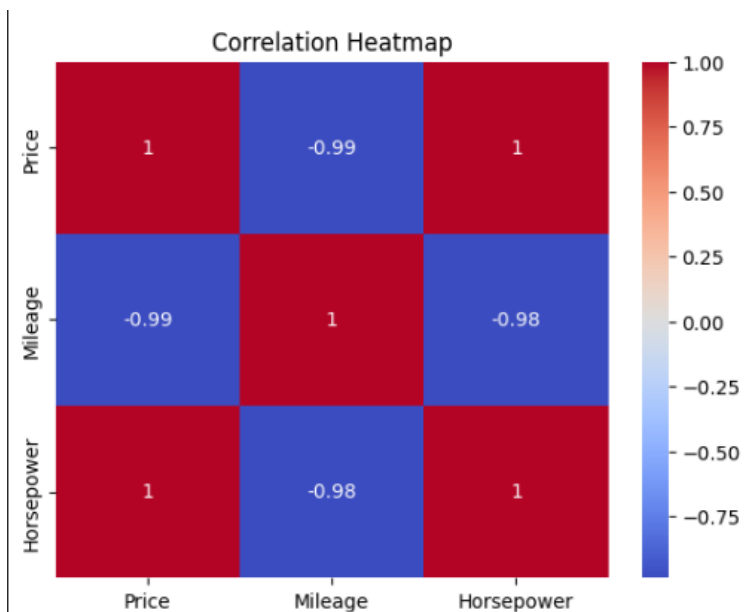
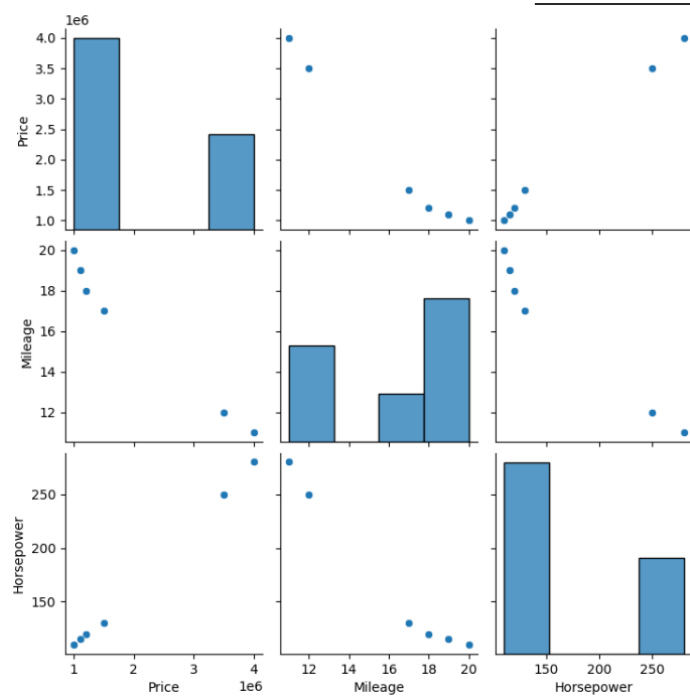
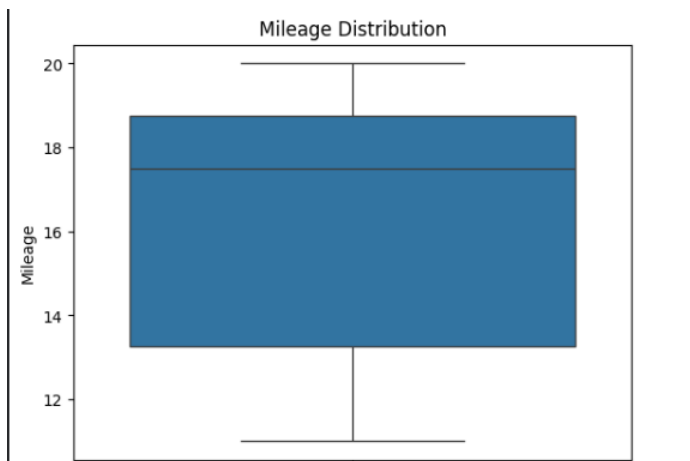
Brand-wise average price Price histogram Mileage boxplot Pairplot Correlation heatmap

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Brand": ["Toyota", "Honda", "BMW", "Toyota", "Honda", "BMW"],
    "Price": [1200000, 1000000, 3500000, 1500000, 1100000, 4000000],
    "Mileage": [18, 20, 12, 17, 19, 11],
    "Horsepower": [120, 110, 250, 130, 115, 280],
    "FuelType": ["Petrol", "Diesel", "Petrol", "Diesel", "Petrol", "Diesel"]
}

df = pd.DataFrame(data)
sns.barplot(data=df, x="Brand", y="Price")
plt.title("Brand-wise Average Price")
plt.show()
sns.histplot(data=df, x="Price", bins=5, kde=True)
plt.title("Price Distribution")
plt.show()
sns.boxplot(data=df, y="Mileage")
plt.title("Mileage Distribution")
plt.show()
sns.pairplot(df[["Price", "Mileage", "Horsepower"]])
plt.show()
sns.heatmap(df[["Price", "Mileage", "Horsepower"]].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```





19.

19. COVID Dataset

Columns

Date Country Confirmed Recovered Deaths

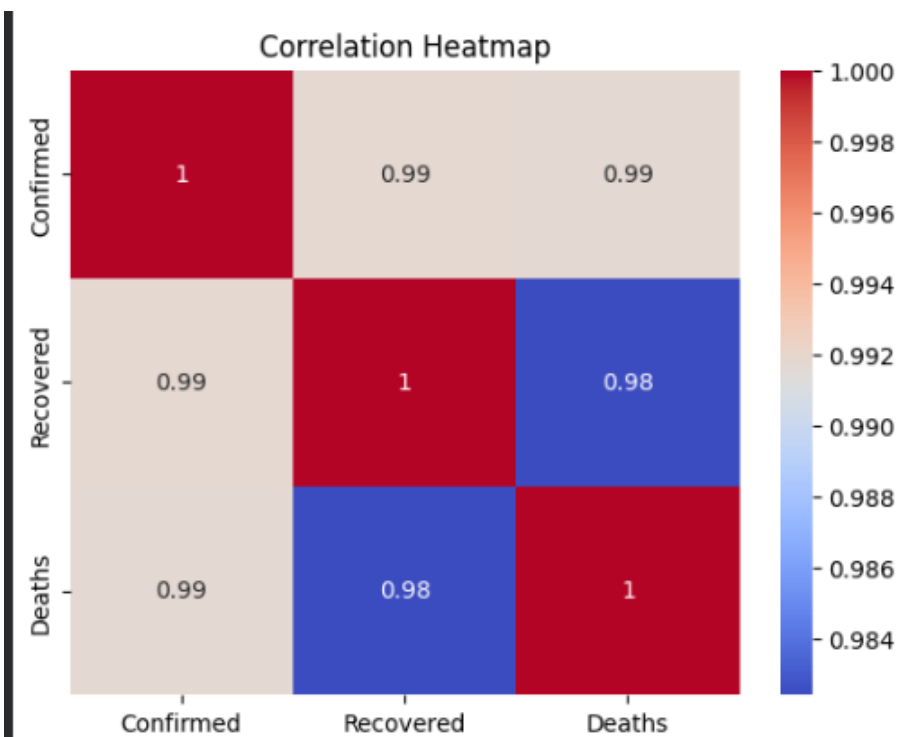
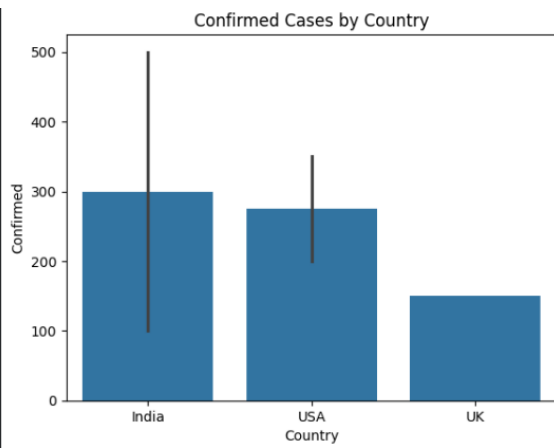
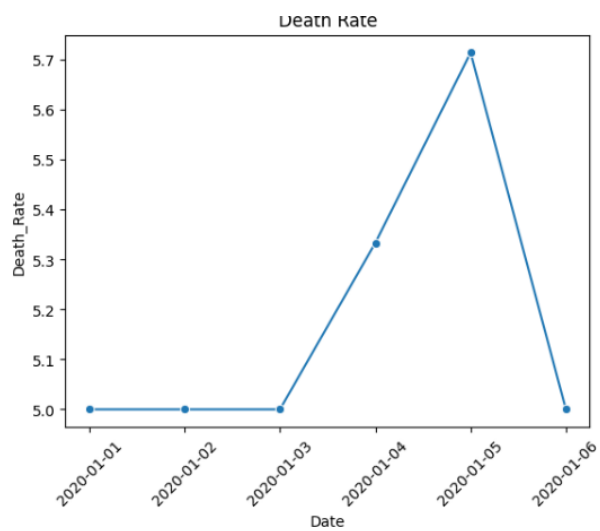
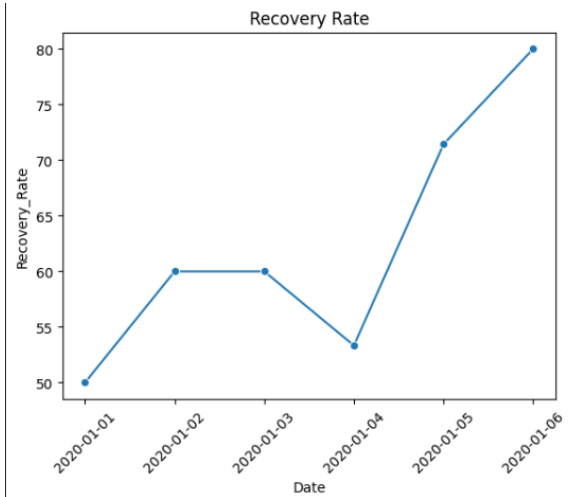
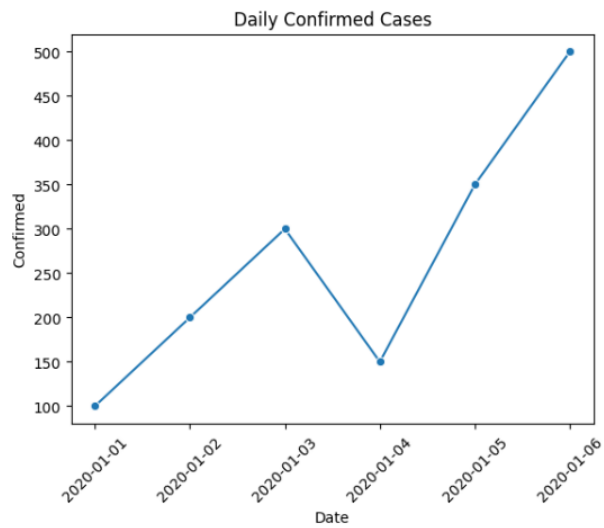
Tasks

Daily trend Recovery rate Death rate Top countries Heatmap

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Date": ["2020-01-01", "2020-01-02", "2020-01-03",
            "2020-01-04", "2020-01-05", "2020-01-06"],
    "Country": ["India", "USA", "India", "UK", "USA", "India"],
    "Confirmed": [100, 200, 300, 150, 350, 500],
    "Recovered": [50, 120, 180, 80, 250, 400],
    "Deaths": [5, 10, 15, 8, 20, 25]
}

df = pd.DataFrame(data)
df["Recovery_Rate"] = (df["Recovered"] / df["Confirmed"]) * 100
df["Death_Rate"] = (df["Deaths"] / df["Confirmed"]) * 100
sns.lineplot(data=df, x="Date", y="Confirmed", marker="o")
plt.title("Daily Confirmed Cases")
plt.xticks(rotation=45)
plt.show()
sns.lineplot(data=df, x="Date", y="Recovery_Rate", marker="o")
plt.title("Recovery Rate")
plt.xticks(rotation=45)
plt.show()
sns.lineplot(data=df, x="Date", y="Death_Rate", marker="o")
plt.title("Death Rate")
plt.xticks(rotation=45)
plt.show()
sns.barplot(data=df, x="Country", y="Confirmed")
plt.title("Confirmed Cases by Country")
plt.show()
sns.heatmap(df[["Confirmed", "Recovered", "Deaths"]].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



20.

20. HR Analytics Dataset (Interview Level)

Columns

EmployeeID Department Salary Experience Age Performance Attrition

Tasks:

Attrition rate Department-wise attrition Salary distribution Experience vs Performance Correlation heatmap Boxplot (Salary by Department) Pairplot Performance histogram Top performers Insights (minimum 10 business insights)

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
data = {
    "EmployeeID": [101,102,103,104,105,106,107,108],
    "Department": ["IT","HR","Finance","IT","Sales","HR","Finance","Sales"],
    "Salary": [50000,40000,60000,70000,45000,42000,65000,48000],
    "Experience": [2,5,8,10,3,6,9,4],
    "Age": [24,30,35,40,27,32,38,29],
    "Performance": [3,4,5,5,3,4,5,4],
    "Attrition": ["No","Yes","No","No","Yes","No","No","Yes"]
}

df = pd.DataFrame(data)
attrition_rate = (df["Attrition"] == "Yes").mean() * 100
print("Attrition Rate:", attrition_rate)
sns.countplot(data=df, x="Department", hue="Attrition")
plt.title("Department-wise Attrition")
plt.show()
sns.histplot(data=df, x="Salary", bins=5, kde=True)
plt.title("Salary Distribution")
plt.show()
sns.scatterplot(data=df, x="Experience", y="Performance", hue="Department", s=100)
plt.title("Experience vs Performance")
plt.show()
sns.heatmap(df[["Salary", "Experience", "Age", "Performance"]].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
sns.boxplot(data=df, x="Department", y="Salary")
plt.title("Salary by Department")
plt.show()
sns.pairplot(df[["Salary", "Experience", "Age", "Performance"]])
plt.show()
sns.histplot(data=df, x="Performance", bins=5)
plt.title("Performance Distribution")
plt.show()
print(df[df["Performance"] == df["Performance"].max()])
```

